

Poisson Point Process - A Model for Random Events

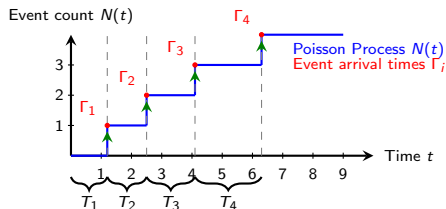
James A. Rose

University of Bristol

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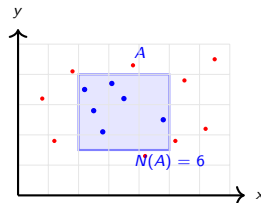
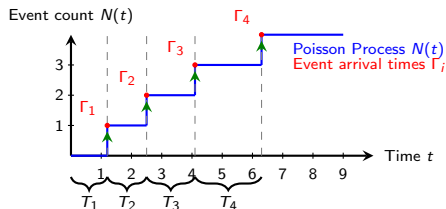
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- Number of events in a fixed time interval is distributed Poisson.
- A Poisson point process is the abstraction of a Poisson process to space.
 - ▶ Instead of events in time, imagine points in space.
 - ▶ The number of points that fall in a set is distributed Poisson
- Examples: arrival time of customers to a cafe, locations of trees in a forest.



History of the Poisson Process

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Hence limit of X as we take $n \rightarrow \infty$ is distributed *Poisson*(λ).

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- Rutherford and others observe properties of radioactive decay and notice:
 - ▶ Decay events happen at 'random' times.
 - ▶ Waiting time between decay events are distributed exponentially.
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These physicists and engineers were observing the Poisson process. 1940s:

- Feller gives official definition for the Poisson process. A Poisson process must have the following properties,
 - 1 independent increments,
 - 2 stationary increments,
 - 3 probability of multiple events in a small interval is negligible - technicality (automatic if waiting times are distributed exponentially).

Poisson Process Definition

There are several equivalent definitions of the Poisson process used today one of which is as follows:

- Let $(X_i)_{i \geq 0}$ be a sequence of random variables such that waiting times $X_i \sim \text{Exponential}(\lambda)$ and $\Gamma_n = \sum_{i=0}^n X_i$. Then

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- Let N be stochastic process. Then N is a *Poisson process with rate λ* if

- 1 $N(0) = 0$,
- 2 N has independent increments, i.e. for any $0 \leq t_0 < \dots < t_k$, the random variables, $N(t_1) - N(t_0), \dots, N(t_n) - N(t_{n-1})$, are independent,
- 3 for any $s, t \geq 0$,

$$N(t + s) - N(s) \sim \text{Poisson}(\lambda t),$$

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- We construct the Poisson process as many Bernoulli arrival 'attempts'. Consider n many intervals of size $h = \frac{t}{n}$, if each interval is distributed approximately Bernoulli:
$$\mathbb{P} \begin{cases} 1 \text{ arrival} \approx \frac{\lambda t}{n} \\ 0 \text{ arrivals} \approx 1 - \frac{\lambda t}{n} \\ > 1 \text{ arrival} \approx 0. \end{cases},$$

then the arrivals are distributed approximately $\text{Binomial}(n, \frac{\lambda t}{n})$ and

$$\text{Binomial}\left(n, \frac{\lambda t}{n}\right) \rightarrow \text{Poisson}(\lambda t).$$

Generalising the Poisson Process

A non-homogeneous Poisson process is a more general form:

- A Poisson process whose rate is not fixed is a non-homogenous Poisson process, its rate is given by the function, μ , the mean measure. Then

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We extend this definition to space:

- Let $\mathcal{E}$ be the class of rectangles in E . Then a point process, N (the function that counts the points in a given set), is a Poisson point process if

- 1 for sets $A \in \mathcal{E}$,

$$N(A) \sim \text{Poisson}(\mu(A)),$$

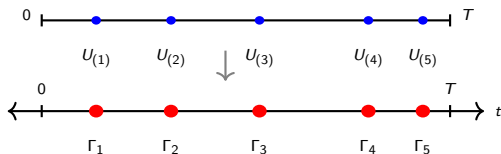
- 2 N is 'completely random', same meaning as independent increments, but for space.

Order Statistic Property

A key property of the Poisson process is that given the number of events, n , in an interval, the distribution of the time of those events is uniform:

- If N is a homogeneous Poisson process on $[0, \infty)$ with rate λ , then

$$(\Gamma_1, \dots, \Gamma_n | N(t) = n) \stackrel{d}{=} (U_{(1)}, \dots, U_{(n)}).$$



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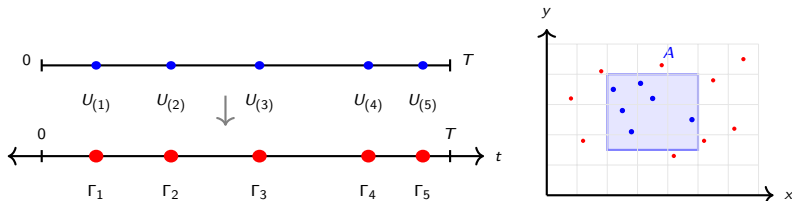
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This property applies to space:

- If mean measure, μ is a multiple of the Lebesgue measure and given the number of points, n , in a set A , then the points in that set are distributed Uniformly.



Applying the Poisson Point Process to the Real World

- Events that are random and that we can assume are independent can be modelled with Poisson process. Like our example of customers arriving to a cafe.
- Items that are distributed randomly in space can be modelled with a Poisson point process.
- We have related this process to the Poisson, Exponential, and Uniform distributions, so we can do lots of calculations about events if we notice they are a Poisson process.
- We can calculate the expectations and probabilities of the number of points. We can talk about how they are spread with the order statistic property.